

1 **Part 5: Properties of MLEs**

2 **Summary**

3 In this Part we explore some of the important properties possessed by max-
4 imum likelihood estimators.

5 Briefly stated, we will see that, asymptotically, these estimators cannot be
6 beat, in the sense that they have minimal MSE, and also are approximatley
7 normally distributed with variance that is easy to calculate. This makes
8 quantifying the error in the estimators straightforward.

1 Invariance of the MLE

2 The invariance result can be stated as follows: If $\hat{\theta}_{\text{MLE}}$ is the MLE for θ , then
3 $g(\hat{\theta}_{\text{MLE}})$ is the MLE for $g(\theta)$.

4 This is a useful, important result because it is often the case that we are
5 truly interested in estimating some function of the parameter θ , not θ itself.

6 Using this, we can say, for example, that if $X_1, X_2, \dots, X_n$ are i.i.d.
7 $\text{normal}(\mu, \sigma^2)$, then the MLE for σ is

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$$\sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

9 and the MLE for the **signal to noise ratio** μ/σ is

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$$\bar{X} / \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

- 1 One consequence of this property is that estimation is not affected by how
2 one chooses to **parameterize** a model.
- 3 For example, some choose to represent the Exponential distribution using
4 the “mean parameterization” which leads to $E(X) = \mu$, whereas we have
5 used the “rate parameterization” which leads to $E(X) = 1/\lambda$. Because
6 of invariance, these two approaches will lead to the same estimates, i.e.,
7 $\hat{\mu} = 1/\hat{\lambda}$.

Exercise: Does unbiasedness have this invariance property?

Exercise: Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Exponential}(\lambda)$. What is the

MLE of the **tail probability** $P(X_i > a)$?

1 Consistency of the MLE

2 An estimator $\hat{\theta}$ is **consistent for θ** if as the sample size n increases, the esti-
3 mator converges to the true value of the parameter.

4 Formally, we would write: For any $\epsilon > 0$,

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$$\lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta_0| > \epsilon) = 0$$

6 This is a fairly low standard placed on an estimator, and, fortunately, under
7 certain “regularity conditions,” the MLE is consistent.

1 Asymptotic Normality of the MLE

2 Another great property of the MLE: The MLE is asymptotically normal,
3 with variance that is not difficult to calculate, under certain “regularity
4 conditions.” (See page 516 from Casella and Berger, Second Edition.)

5 This will allow an easy route to:

6 1. Calculate standard errors for MLEs

7 2. Constructing confidence intervals for parameters

8 3. Constructing confidence intervals for general functions of parameters
9 (via the Delta method)

10 4. Constructing hypothesis tests for parameters

- 1 To start, we focus on the case where the parameter θ is one-dimensional.
- 2 The formal result can be stated as follows:
- 3 If $X_1, X_2, \dots, X_n$ are i.i.d. $f(x; \theta_0)$, then, under suitable regularity condi-
- 4 tions, as n increases,

$$\sqrt{n} (\hat{\theta} - \theta_0) \longrightarrow N(0, I^{-1}(\theta_0))$$

6 where

$$I(\theta_0) = E\left(-\frac{\partial^2 \log f(X; \theta)}{\partial \theta^2}\right)$$

8 is the **Fisher Information**.

1 **Exercise:** Explain the practical value of the above result.

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1 **Exercise:** Show that

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$$nI(\theta) = E \left[-\frac{\partial^2}{\partial \theta^2} \log L(\theta) \right]$$

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- 1 The previous result shows that the variance of the MLE is approximately
2 equal to the reciprocal of the expected value of the negative second deriva-
3 tive of the log likelihood at its peak.
- 4 And, “the negative of the second derivative” is the amount of curvature at
5 the peak of the likelihood.
- 6 If the log likelihood has a “sharp” peak, then $nI(\theta)$ will be large, and the
7 variance of $\hat{\theta}$ will be small.

1 **Exercise:** Assume $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, where σ^2 is known. We
2 know that $\bar{X}$ is the MLE for μ . Derive the Fisher Information for μ , and use
3 it to approximate the standard error and distribution of $\bar{X}$.

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- 1 **Exercise:** Assume that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Exponential}(\lambda)$. Derive the
- 2 Fisher Information for λ and the approximate distribution of its MLE.

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- 1 **Exercise:** Assume that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Poisson}(\lambda)$. Derive the
2 Fisher Information for λ and the approximate distribution of its MLE.

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1 Efficiency of the MLE

2 It is possible to show that the lower bound on the variance of any unbiased
3 estimator is $1/nI(\theta_0)$. This is called the **Cramer-Rao lower bound**.

4 Since the MLE is asymptotically unbiased, we know that we are achieving
5 the Cramer-Rao lower bound, at least asymptotically.

6 Hence, we are minimizing the MSE (asymptotically). This is objectively
7 the best rationale for utilizing maximum likelihood estimators.

1 The Delta Method

2 Often, our objective is not to estimate θ , but instead some function of θ , call
3 it $g(\theta)$. For example, often want to estimate the probability of some event
4 A . This probability is not a parameter, but it is a function of θ .

5 From the invariance property we know that the MLE of $g(\theta)$ is $g(\hat{\theta})$. But,
6 we also need to calculate a SE or construct a confidence interval for this
7 estimate.

8 One approach would be to “reparameterize” the model in terms of $g(\theta)$,
9 and then find $I(g(\theta))$, and then use $1/I(g(\theta))$ as the variance, etc. Fortu-
10 nately, there is an easier way...

1 Recall the **Delta Method**:

2 If Y_n is approximately $N(\mu, \sigma_n^2)$ and $\sigma_n^2 \rightarrow 0$ as n increases, then
3 $g(Y_n)$ is approximately $N(g(\mu), (g'(\mu))^2 \sigma_n^2)$, assuming that $g'(\mu)$ ex-
4 ists and is not zero.

5 Here we will use this result with the MLE for θ in place of Y_n .

6 Then, $\mu = \theta_0$ and $\sigma_n^2 = 1/nI(\theta_0)$.

7 We can conclude that $g(\hat{\theta})$ is approximately

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$$N\left(g(\theta_0), g'(\theta_0)^2 \left(\frac{1}{nI(\theta_0)}\right)\right)$$

9 In practice, we use the approximate variance:

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$$g'(\hat{\theta})^2 \left(\frac{1}{nI(\hat{\theta})}\right)$$

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Exponential}(\lambda)$. We seek to
2 estimate $\tau = P(X_i > a)$ for some constant $a > 0$. Construct the MLE and
3 find its standard error. Also construct a confidence interval for τ .

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1 The Multidimensional Case

2 Now we consider the case where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_p)$.

3 We define the **Fisher Information matrix** to be the p by p matrix whose (i, j)
4 entry is

$$5 \quad I_{ij}(\boldsymbol{\theta}) = E \left[-\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(x; \boldsymbol{\theta}) \right]$$

6 **The Main Result:** Under appropriate “regularity conditions,” if $X_1, X_2, \dots, X_n$
7 are i.i.d. $f(x; \boldsymbol{\theta})$ then

$$8 \quad \sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \longrightarrow N_p(0, \mathbf{I}^{-1}(\boldsymbol{\theta}_0))$$

9 Recall that N_p denotes the p -dimensional multivariate normal distribution.

10 **Practical Version:** Under “regularity conditions,” $\hat{\boldsymbol{\theta}}$ is approximately nor-
11 mal with variance $\mathbf{I}^{-1}(\hat{\boldsymbol{\theta}})/n$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$. Derive the joint
2 distribution of the MLE for $\boldsymbol{\theta} = (\mu, \sigma^2)$.

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1 Multivariate Delta Method

2 We can extend the Delta method to cases where the parameter is multidimensional.
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4 In general, suppose that $\mathbf{Y}_n$ is approximately $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}/n)$. Now consider
5 an m -dimensional function $\mathbf{g}()$ where $m \leq p$:

$$\mathbf{g}(\mathbf{Y}_n) = \begin{bmatrix} g_1(\mathbf{Y}_n) \\ g_2(\mathbf{Y}_n) \\ \vdots \\ g_m(\mathbf{Y}_n) \end{bmatrix}$$

7 Then, $\mathbf{g}(\mathbf{Y}_n)$ is approximately $N_m(\mathbf{g}(\boldsymbol{\mu}), \mathbf{G}\boldsymbol{\Sigma}\mathbf{G}^T/n)$ where the (i, j) entry of
8 $\mathbf{G}$ is $\partial g_i(\mathbf{y})/\partial y_j$. This all assumes that $\mathbf{G}$ is of full rank.

- 1 **Exercise:** Place our parameter estimation problem into the framework of
2 the above statement of the Delta method.

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1 **Case Study: Geometric Brownian Motion**

- 2 This Part concludes with a simple, but important, example of how careful
- 3 derivation of the MLE and its distribution can lead to useful insight.
- 4 We begin by defining some standard concepts regarding Brownian motion.

1 *The Standard Brownian Motion (Wiener Process)*

2 A stochastic process $\{W(t) : t \geq 0\}$ is a **(standard) Brownian motion** if the
3 following hold:

- 4 1. $W(0) = 0$. (almost surely)
- 5 2. $W(t)$ is continuous as a function of t . (almost surely)
- 6 3. $W(t_4) - W(t_3)$ and $W(t_2) - W(t_1)$ are independent for $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 7 4. The distribution of $W(t_2) - W(t_1)$ depends only on $|t_1 - t_2|$ for t_1, t_2 .
- 8 5. $W(t)$ is $\text{Normal}(0, t)$ for all t .

1 *Brownian Motion with Drift and Scaling*

2 A stochastic process $\{B(t) : t \geq 0\}$ is a **Brownian Motion with drift μ and**
3 **scaling σ^2** if $\{W(t) : t \geq 0\}$ is a standard Brownian motion and

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$$B(t) = \mu t + \sigma W(t).$$

5 Note that this implies the following:

- 6 1. $B(0) = 0$. (almost surely)
- 7 2. $B(t)$ is continuous as a function of t . (almost surely)
- 8 3. $B(t_4) - B(t_3)$ and $B(t_2) - B(t_1)$ are independent for $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 9 4. The distribution of $B(t_2) - B(t_1)$ depends only on $|t_1 - t_2|$ for t_1, t_2 .
- 10 5. $B(t)$ is $\text{Normal}(\mu t, \sigma^2 t)$ for all t .

Geometric Brownian Motion

A stochastic process $\{S(t) : t \geq 0\}$ is a **geometric Brownian motion** if $\{B(t) : t \geq 0\}$ is a Brownian motion with drift μ and scaling σ^2 and

$$S(t) = S(0) e^{B(t)}$$

where $S(0) > 0$ is the initial value. Define $\alpha = \mu + \sigma^2/2$.

Note that this implies the following:

1. $S(t)$ is continuous as a function of t . (almost surely)
2. $S(t_4)/S(t_3)$ and $S(t_2)/S(t_1)$ are independent for any $t_1 \leq t_2 \leq t_3 \leq t_4$.
3. The distribution of $S(t_2)/S(t_1)$ depends only on $|t_1 - t_2|$ for any t_1, t_2 .
4. $S(t)$ is Lognormal($\log(S(0)) + \mu t, \sigma^2 t$) for all t .
5. $S(t)$ has expected value $S(0)e^{\alpha t}$.
6. $\log(S(t_2)/S(t_1))$ is Normal($\mu(t_2 - t_1), \sigma^2(t_2 - t_1)$) for all $t_1 \leq t_2$.

- 1 Let's consider the geometric Brownian motion model. Can we use data
2 (observed prices) to estimate α and σ^2 ?
- 3 Suppose we observe the stock price over the period $[0, T]$. We will model
4 the price as a geometric Brownian motion $\{S(t) : t \geq 0\}$.
- 5 Divide the time interval into n equal-width subintervals, with $\Delta = T/n$.
- 6 Define, for $i = 1, 2, \dots, n$,
- 7
$$Y_i = \log \left(\frac{S(t_{i-1} + \Delta)}{S(t_{i-1})} \right)$$
- 8 where $t_i = i/n$, the times at which the price is observed.
- 9 From Property 6 of a geometric Brownian motion, $Y_1, Y_2, \dots, Y_n$ are i.i.d.
10 $N(\Delta\mu, \Delta\sigma^2)$.

1 **Exercise:** Find the MLE of $\theta = (\alpha, \sigma^2)$. Approximate the distribution of θ .

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- 1 **Exercise:** What does this result suggest regarding how large n should be
2 chosen for the best estimation of σ^2 ? What about for α ?

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- 1 Is it, in fact, a good idea to sample more frequently in order to estimate
2 σ^2 ? See Ait-Sahalia, Mykland, and Zhang (2005), “How Often to Sam-
3 ple a Continuous-Time Process in the Presence of Market Microstructure
4 Noise,” *The Review of Financial Studies*, Volume 18.